Ranjan Singh

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Links: GitHub | Portfolio | LinkedIn | LeetCode | GFG

Summary

Computer Science undergraduate passionate about software development, with hands-on experience in full-stack engineering. Seeking an opportunity to apply my technical skills to deliver practical, real-world solutions.

Education

(2023 - 2027) | B.Tech, Computer Science & Engineering | Maharaja Agrasen Institute of Technology, Delhi |
CGPA: 8.6

Trainings / Certifications

Oracle AI Foundations Associate, Full Stack Web Development Bootcamp (Udemy), Introduction to Graph Algorithms (SWAYAM/NPTEL)
Open-Source Contributions- Contributed in PingUp in GSSOC 2026 - GitHub Link

Technical Skills

Languages: C, C++, Java, Python, HTML, CSS, JavaScript, TypeScript
Libraries/Frameworks: React.js, Next.js, Node.js, Express.js, FastAPI, Tailwind CSS, Redux/Zustand, Framer Motion
Databases & Backend Services: MongoDB, PostgreSQL, Supabase, Appwrite
Tools & Platforms: Git, GitHub, VS Code, Postman/ThunderClient, Docker, AWS, Linux

Projects

• **SnapBuy – Multi-Vendor Ecommerce Platform | GitHub | Live Demo**
Developed a production-style full-stack ecommerce platform using React, FastAPI, MongoDB, Razorpay, and Cloudinary.
– Implemented secure authentication with JWT, HTTP-only cookies, and role-based access for customers and merchants.
– Built complete ecommerce workflows including product management, search and category-based filtering, dashboard features, role-based user workflows and features, checkout, and order tracking.

• **HealthCare Platform Architecture – Asynchronous Microservices Backend** | GitHub
Developed a robust, containerized booking and payment processing backend for diagnostic centres using FastAPI, PostgreSQL, Redis, and Celery.
– System Architecture: Orchestrated a 4-container deployment (FastAPI web server, PostgreSQL database, Redis broker, Celery worker) via Docker Compose on a private virtual network with native health checks.
– Asynchronous Webhooks: Engineered a non-blocking payment webhook pipeline that instantly queues incoming event payloads in Redis, allowing the API to handle high concurrent traffic while the Celery worker executes database updates in the background.
– Fault Tolerance & Idempotency: Guaranteed 100% data consistency by implementing a PostgreSQL transaction ledger to block duplicate webhook events, and configured Celery with exponential backoff retries to recover gracefully from database locks or network volatility.
– Security & Validation: Secured endpoints with stateless JWT OAuth2 authentication and IP-based rate limiting (slowapi) to mitigate DDoS attacks. Enforced strict server-side relational price computations via SQLAlchemy to prevent client-side payload manipulation.

– Optimization & Testing**:** Implemented cursor-based pagination for memory-safe database queries and wrote a comprehensive Pytest suite utilizing an isolated SQLite memory instance and synchronous Celery execution to test relational logic without cross-contamination.

• **Blogify – Full-Stack Blogging Platform | GitHub | Live Demo**

Developed a full-stack blogging web application using React, Tailwind CSS, and Appwrite, enabling users to create, publish, and interact with blogs.

– Implemented role-based authentication and protected routes.

– Integrated blog & comment CRUD with markdown editor and image upload.

• **Taskin –Task Management App | GitHub | Live Demo**

Developed a full-stack task management web app using React, Node.js, Express, and MongoDB.

– Implemented user authentication, CRUD operations, task categorization, analytics dashboard, and calendar integration.

– Added search, sort, and filter features for improved task organization.

Achievements

– Top 30 out of 700+ teams at GGSIPU Vibestate Hackathon (2025) reached the semi-finals for building MindMateAI app.

– Reached the semi-finals in ET GenAI Hackathon 2026 with ETNewsSphere (Multi AI-Agent-Based Project) and was recognized among the top-performing candidates.

– Solved 400+ Data Structures & Algorithms problems across LeetCode and GeeksforGeeks.

– Ranked among top 26% in HackerRank Orchestrate 24-hour Hackathon (June 2026 Edition).